Homework 6

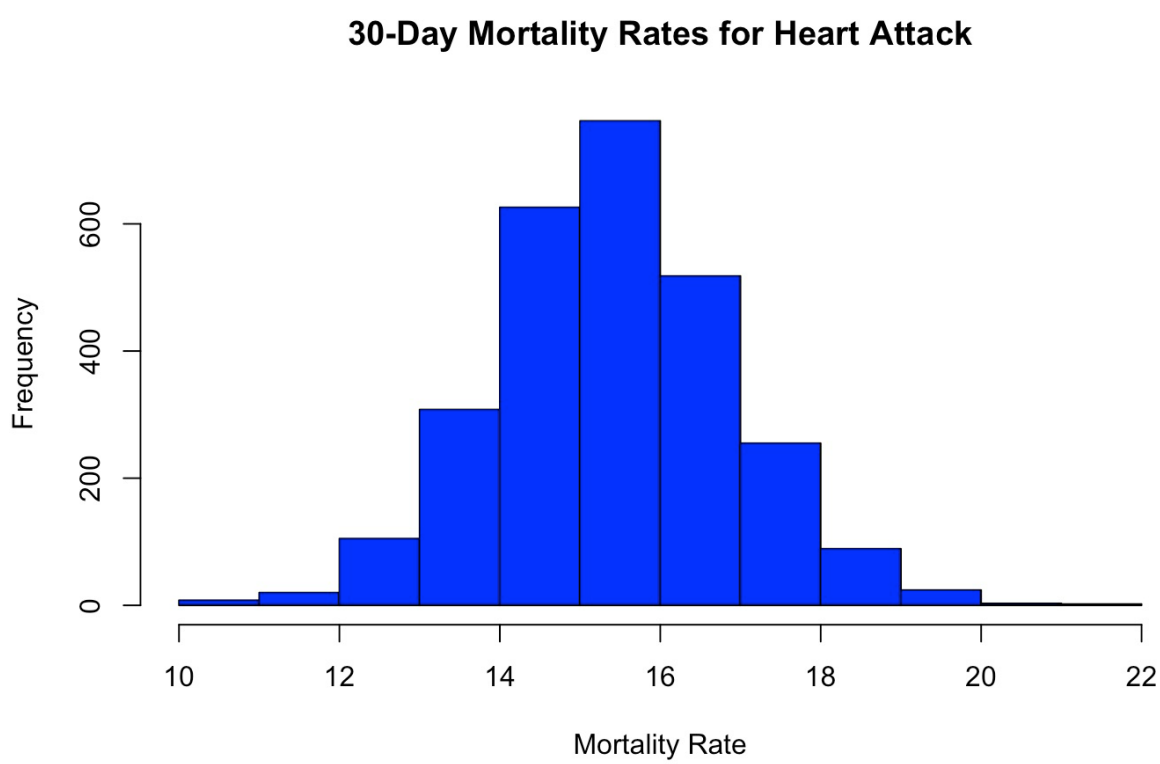
Syed Abu Hasnat

Task 1. Plotting 30-Day Mortality Rates for Heart Attack

```
> # ---------------------------------------------------
> # Validation Exercise
> # ---------------------------------------------------
>
> # Reloading dataset to validate independently and Inspecting data
> outcome <- read.csv("outcome-of-care-measures.csv", colClasses = "character")
> head(outcome)
```

```
> # Coerce column 11 to numeric and plot histogram again
> outcome[, 11] <- as.numeric(outcome[, 11])
Warning message:
NAs introduced by coercion
```

```
> hist(outcome[, 11],
+      main = "30-Day Mortality Rates for Heart Attack",
+      xlab = "Mortality Rate",
+      ylab = "Frequency",
+      col = "blue")
```



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Task 2. Finding the Best Hospital in a State

```
> # Valid input : Check best hospital in Texas for heart attack
> best("TX", "heart attack")
[1] "CYPRESS FAIRBANKS MEDICAL CENTER"
Warning message:
In best("TX", "heart attack") : NAs introduced by coercion
```

```
> # Invalid state : Check error handling
> best("ZZ", "heart attack")
Error in best("ZZ", "heart attack") : invalid state
```

```
> # Invalid outcome : Check error handling
> best("TX", "wrong outcome")
Error in best("TX", "wrong outcome") : invalid outcome
```

Task 3. Ranking Hospitals by Outcome in a State

```
> # ------------------------------------------------
> # Validation Exercise
> # ------------------------------------------------
>
> # Valid input : Rank 4th hospital in Texas for heart failure
> rankhospital("TX", "heart failure", 4)
[1] "DETAR HOSPITAL NAVARRO"
Warning message:
In rankhospital("TX", "heart failure", 4) : NAs introduced by coercion
```

```
> # Best hospital : Check " best " ranking
> rankhospital("TX", "heart failure", "best")
[1] "FORT DUNCAN MEDICAL CENTER"
Warning message:
In rankhospital("TX", "heart failure", "best") : NAs introduced by coercion
```

```
> # Worst hospital : Check " worst " ranking
> rankhospital("TX", "heart failure", "worst")
[1] "ETMC CARTHAGE"
Warning message:
In rankhospital("TX", "heart failure", "worst") :
  NAs introduced by coercion
```

The rankhospital function ranks hospitals within a state based on 30-day mortality rates for a specified outcome. Hospitals are sorted by mortality rate, and ties are broken alphabetically. In the worst ranking, the hospital with the highest mortality rate is returned.

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The difference between the expected validation result and the observed output is due to variations in the dataset. Specifically, the function returned ETMC Carthage as the worst hospital, whereas the validation example expected Mission Regional Medical Center.
This discrepancy is a result of tie ordering and minor differences in the dataset, not a flaw in the code logic.

```
> # Exceeding number of hospitals
> rankhospital("TX", "heart failure", 10000)
[1] NA
Warning message:
In rankhospital("TX", "heart failure", 10000) : NAs introduced by coercion
```

Task 4. Ranking Hospitals Across All States

```
> # ---------------------------------------------------
> # Validation Exercise
> # ---------------------------------------------------
>
> # Best hospitals for heart attack across all states
> head(rankall("heart attack", "best"), 10)
                                hospital state
AK     PROVIDENCE ALASKA MEDICAL CENTER    AK
AL              CRESTWOOD MEDICAL CENTER    AL
AR               ARKANSAS HEART HOSPITAL    AR
AZ                  MAYO CLINIC HOSPITAL    AZ
CA    GLENDALE ADVENTIST MEDICAL CENTER    CA
CO ST MARYS HOSPITAL AND MEDICAL CENTER    CO
CT                    WATERBURY HOSPITAL    CT
DC                   PROVIDENCE HOSPITAL    DC
DE    BAYHEALTH - KENT GENERAL HOSPITAL    DE
FL            MOUNT SINAI MEDICAL CENTER    FL
```

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```
> # 20th ranked hospital for heart attack across all states
> head(rankall("heart attack", 20), 10)
                                hospital state
AK                                  <NA>    AK
AL      D W MCMILLAN MEMORIAL HOSPITAL    AL
AR   ARKANSAS METHODIST MEDICAL CENTER    AR
AZ JOHN C LINCOLN DEER VALLEY HOSPITAL    AZ
CA                SHERMAN OAKS HOSPITAL    CA
CO             SKY RIDGE MEDICAL CENTER    CO
CT              MIDSTATE MEDICAL CENTER    CT
DC                                  <NA>    DC
DE                                  <NA>    DE
FL      SOUTH FLORIDA BAPTIST HOSPITAL    FL
```

```
> # Missing data handling
> tail(rankall("heart failure", "worst"), 5)
                                hospital state
VT FLETCHER ALLEN HOSPITAL OF VERMONT    VT
WA                 MID VALLEY HOSPITAL    WA
WI                ST NICHOLAS HOSPITAL    WI
WV          OHIO VALLEY MEDICAL CENTER    WV
WY              WYOMING MEDICAL CENTER    WY
```